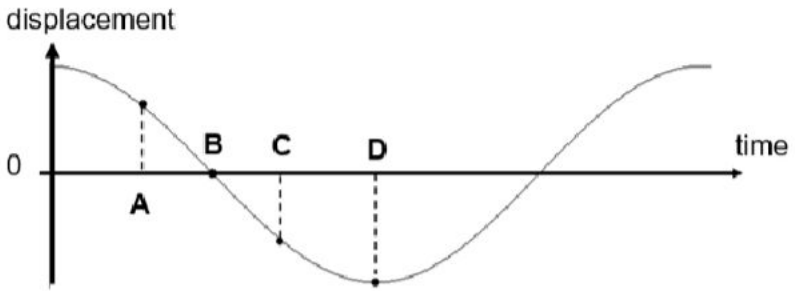
9 The diagram shows the graph of displacement against time for a body performing simple harmonic motion. At which time, **A, B, C** or **D** is the body travelling and accelerating in the opposite direction?



Ans: C

The velocity-time graph of the above motion will be a negative sine graph and the acceleration-time graph will be a negative cosine graph.

	A	B	C	D
displacement	+	0	-	-
velocity	-	-	-	0
acceleration	-	0	+	+

Option C shows that the body is traveling (which is shown by the direction of velocity) and accelerating in opposite direction.